

```
#Part 1: Environment Setup and Tool Overview
# Install required libraries (uncomment if needed)
# !pip install pandas numpy matplotlib scikit-learn

# Import libraries with standard aliases
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn import datasets
import warnings
warnings.filterwarnings('ignore') # Hide warning messages for cleaner o

print("✅ All libraries imported successfully!")
print(f"Pandas version: {pd.__version__}")
print(f"NumPy version: {np.__version__}")
```

```
✅ All libraries imported successfully!
Pandas version: 2.2.3
NumPy version: 2.1.3
```

```
#Part 2: Loading and Exploring Your First Dataset
# Load a simple dataset (Iris flowers – a classic beginner dataset)
from sklearn.datasets import load_iris

# Load the data
iris = load_iris()
print("Dataset loaded successfully!")
print(f"Dataset shape: {iris.data.shape}")
print(f"Features: {iris.feature_names}")
print(f"Target classes: {iris.target_names}")
```

```
Dataset loaded successfully!
Dataset shape: (150, 4)
Features: ['sepal length (cm)', 'sepal width (cm)', 'petal length (cm)',
Target classes: ['setosa' 'versicolor' 'virginica']
```

```
# Convert to pandas DataFrame for easier handling
df = pd.DataFrame(iris.data, columns=iris.feature_names)
df['species'] = iris.target_names[iris.target]

# Display first few rows
print("First 5 rows of our dataset:")
print(df.head())

print("\nDataset info:")
print(df.info())
```

First 5 rows of our dataset:

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)
0	5.1	3.5	1.4	0.4
1	4.9	3.0	1.4	0.4
2	4.7	3.2	1.3	0.4
3	4.6	3.1	1.5	0.4
4	5.0	3.6	1.4	0.5

```
species
0  setosa
1  setosa
2  setosa
3  setosa
4  setosa
```

Dataset info:

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 150 entries, 0 to 149
```

```
Data columns (total 5 columns):
```

#	Column	Non-Null Count	Dtype
0	sepal length (cm)	150 non-null	float64
1	sepal width (cm)	150 non-null	float64
2	petal length (cm)	150 non-null	float64
3	petal width (cm)	150 non-null	float64
4	species	150 non-null	object

```
dtypes: float64(4), object(1)
```

```
memory usage: 6.0+ KB
```

```
None
```

#Part 3: Creating Your First Visualization

Create a simple scatter plot

```
plt.figure(figsize=(10, 6))
```

Plot sepal length vs sepal width, colored by species

```
species_colors = {'setosa': 'red', 'versicolor': 'blue', 'virginica': 'green'}
```

```
for species in df['species'].unique():
```

```
    species_data = df[df['species'] == species]
```

```
    plt.scatter(species_data['sepal length (cm)'],
```

```
                species_data['sepal width (cm)'],
```

```
                c=species_colors[species],
```

```
                label=species,
```

```
                alpha=0.7)
```

```
plt.xlabel('Sepal Length (cm)')
```

```
plt.ylabel('Sepal Width (cm)')
```

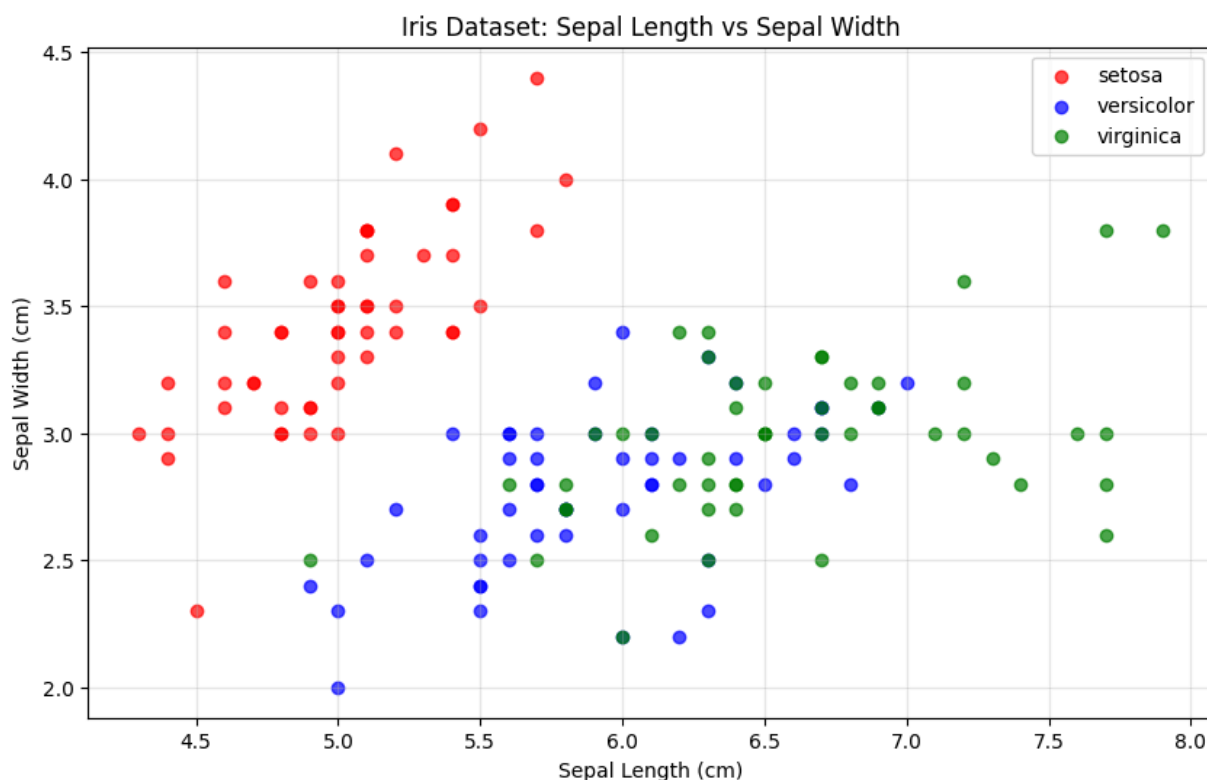
```
plt.title('Iris Dataset: Sepal Length vs Sepal Width')
```

```
plt.legend()
```

```
plt.grid(True, alpha=0.3)
```

```
plt.show()
```

print("🎉 Congratulations! You've created your first data visualization!")



🎉 Congratulations! You've created your first data visualization!

#Part 4: Practice with Basic Data Operations

Basic statistical analysis

print("Basic Statistics for Iris Dataset:")

print("=" * 40)

Calculate mean values for each species

species_means = df.groupby('species').mean()

print("\nMean values by species:")

print(species_means)

Count samples per species

species_counts = df['species'].value_counts()

print("\nSamples per species:")

print(species_counts)

Basic Statistics for Iris Dataset:

=====

Mean values by species:

	sepal length (cm)	sepal width (cm)	petal length (cm)	\
species				
setosa	5.006	3.428	1.462	
versicolor	5.936	2.770	4.260	
virginica	6.588	2.974	5.552	

	petal width (cm)
species	
setosa	0.246
versicolor	1.326
virginica	2.026

Samples per species:

species	
setosa	50
versicolor	50
virginica	50

Name: count, dtype: int64

Double-click (or enter) to edit

```
#Assessment: Tool Familiarity Check
# Task 1: Create a simple calculation using NumPy
# Calculate the mean and standard deviation of sepal length

sepal_lengths = df['sepal length (cm)']

# Your code here:
mean_sepal_length = np.mean(sepal_lengths)
std_sepal_length = np.std(sepal_lengths)

print(f"Mean sepal length: {mean_sepal_length:.2f} cm")
print(f"Standard deviation: {std_sepal_length:.2f} cm")

# Verification (don't modify)
assert isinstance(mean_sepal_length, (float, np.floating)), "Mean should
assert isinstance(std_sepal_length, (float, np.floating)), "Std should b
print("✅ Task 1 completed successfully!")
```

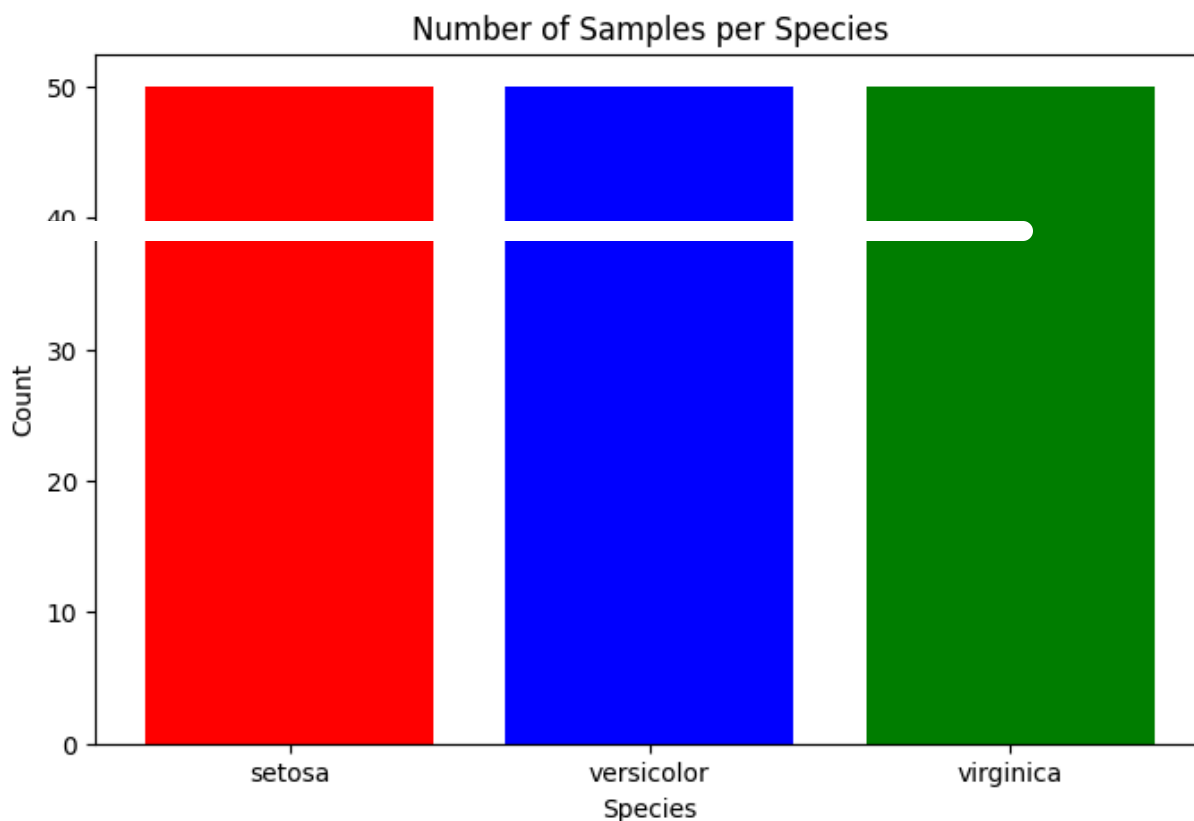
Mean sepal length: 5.84 cm
 Standard deviation: 0.83 cm
 ✅ Task 1 completed successfully!

```
# Task 2: Create a simple bar chart showing species counts
species_counts = df['species'].value_counts()

plt.figure(figsize=(8, 5))
plt.bar(species_counts.index, species_counts.values, color=['red', 'blue
plt.title('Number of Samples per Species')
```

```
plt.xlabel('Species')
plt.ylabel('Count')
plt.show()

print(f"Species distribution: {dict(species_counts)}")
print("✅ Task 2 completed successfully!")
```



Species distribution: {'setosa': np.int64(50), 'versicolor': np.int64(50), 'virginica': np.int64(50)}

✅ Task 2 completed successfully!

Analysis and Reflection

My Observations About the Iris Dataset

Dataset Overview:

Number of samples: 150

Number of features: 4

Number of classes: 3

Key Findings from the Visualization:

The scatter plot shows that the different Iris species form somewhat distinct groups based on sepal length and sepal width. Some species are more clearly separated than others, while there is some overlap between the data points. The visualization shows that

sepal measurements can help distinguish between Iris species, although they may not be enough to perfectly separate all three classes.

Questions for Further Investigation:

1. Which of the four features is the best for distinguishing between the three Iris species?
2. Would a machine learning classification model be able to accurately predict the species using these features?

Reflection:

I learned how to use Python libraries such as Pandas, NumPy, Matplotlib, and Scikit-learn to load, analyze, and visualize a dataset. I also learned how visualizations and basic statistical analysis can help us understand patterns in data before applying machine learning models.